

KONOTOP, Ukraine

WMO: 332610

Lat: 51.2408N

Lon: 33.1869E

Elev: 489

StdP: 14.44

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -3.3	(c) 1.9	(d) -9.1	(e) 3.4	(f) -2.4	(g) -4.2	(h) 4.5	(i) 2.7	(j) 16.0	(k) 19.8	(l) 14.3	(m) 21.7	(n) 3.4	(o) 50	(p) 0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	88.2	68.7	85.0	67.4	82.0	66.0	71.0	83.4	69.5	81.2	68.0	78.6	5.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.0	101.2	76.4	65.5	95.9	74.5	64.1	91.1	73.2	35.1	83.7	33.8	81.3	32.6	79.0	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 13.8	(b) 12.1	(c) 10.1	(e) -8.6	(f) 92.8	(g) 5.9	(h) 3.6	(i) -12.8	(j) 95.4	(k) -16.3	(l) 97.5	(m) -19.7	(n) 99.5	(o) -24.0	(p) 102.1		
(a) 13.8	(b) 12.1	(c) 10.1	(e) -9.0	(f) 73.6	(g) 5.9	(h) 2.1	(i) -13.3	(j) 75.1	(k) -16.7	(l) 76.3	(m) -20.0	(n) 77.4	(o) -24.3	(p) 78.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	23.0	24.9	33.8	48.7	59.7	65.9	69.4	67.2	57.1	45.5	35.5	26.6		
	DBStd	18.57	10.32	10.65	8.31	7.96	8.25	6.52	5.55	6.37	7.32	7.87	8.45	10.12		
	HDD50	3545	838	703	505	119	15	0	0	19	183	437	726			
	HDD65	7164	1303	1123	969	491	206	70	21	48	253	605	884	1191		
	CDD50	2284	0	0	1	78	316	477	600	534	232	43	3	0		
	CDD65	428	0	0	0	1	41	96	156	117	17	0	0	0		
	CDH74	3799	0	0	0	16	388	777	1293	1150	174	1	0	0		
	CDH80	1094	0	0	0	2	85	186	390	398	33	0	0	0		
(14) Wind	WSAvg	4.7	5.4	5.4	5.7	5.1	4.5	4.3	3.7	3.7	3.8	4.3	5.2	5.4		
(15) Precipitation	PrecAvg	23.7	1.8	1.5	1.6	1.6	2.2	2.5	2.8	2.1	2.1	1.9	1.8	1.7		
	PrecMax	34.1	3.1	4.0	4.8	3.7	5.3	5.6	7.8	6.0	5.7	4.1	3.8	3.8		
	PrecMin	18.3	0.5	0.3	0.0	0.1	0.8	0.7	0.7	0.1	0.1	0.1	0.3	0.3		
	PrecStd	3.8	0.7	0.7	1.0	0.9	1.1	1.1	1.6	1.4	1.4	1.1	0.9	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.9	45.7	59.8	76.1	86.6	88.5	92.5	94.5	84.1	72.0	58.0	48.0		
		MCWB	41.0	41.7	46.9	56.9	64.7	67.5	70.2	68.5	63.9	59.1	54.1	46.2		
	2%	DB	39.3	41.1	53.6	70.6	82.3	84.9	87.8	88.8	79.2	66.0	52.6	44.0		
		MCWB	37.5	38.9	44.4	54.7	62.9	67.5	69.3	69.3	62.1	56.9	49.5	42.5		
	5%	DB	36.8	38.3	48.9	66.3	78.5	81.8	84.6	84.2	74.4	60.7	49.2	40.4		
		MCWB	35.2	36.4	42.2	53.0	61.9	66.0	68.4	67.1	60.4	54.7	47.1	39.2		
	10%	DB	34.8	36.3	44.8	62.4	74.5	78.8	81.1	80.3	69.9	56.6	46.6	37.8		
		MCWB	33.5	34.7	39.8	51.0	60.3	65.0	67.3	65.2	58.7	51.7	44.8	36.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.4	42.4	49.3	59.2	67.9	71.3	73.3	72.7	66.8	61.8	54.1	46.5		
		MCDB	42.6	44.7	57.9	70.8	80.6	81.6	85.4	85.6	78.8	68.7	57.8	47.9		
	2%	WB	37.7	38.8	45.5	56.1	65.4	69.2	71.5	70.6	63.7	58.0	50.2	42.6		
		MCDB	38.9	40.5	51.3	66.2	77.6	80.7	83.8	84.2	75.5	63.9	52.0	43.8		
	5%	WB	35.5	36.7	42.8	54.3	63.5	67.6	70.0	68.7	61.7	55.1	47.3	39.4		
		MCDB	36.6	38.0	48.4	63.6	74.7	78.7	81.3	81.3	71.0	60.1	48.8	40.3		
	10%	WB	33.7	35.0	40.3	52.4	61.7	66.1	68.5	66.9	59.8	52.6	44.9	36.7		
		MCDB	34.6	36.1	44.5	60.9	71.9	76.3	78.4	77.6	67.8	55.9	46.5	37.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	9.3	11.9	16.4	18.7	17.2	17.5	19.5	17.1	13.5	7.9	7.3		
		MCDBR	6.2	8.7	18.2	22.2	23.2	21.4	21.9	25.3	24.3	20.3	10.7	7.6		
		MCWBR	5.7	7.3	11.1	10.5	8.9	8.5	7.9	9.1	10.7	11.8	8.4	6.9		
	5% WB	MCDBR	6.2	8.2	16.1	19.0	20.4	18.9	19.3	22.4	20.9	17.7	9.4	7.1		
		MCWBR	5.8	7.3	10.7	9.8	9.1	8.7	8.1	9.0	10.5	11.4	8.1	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.306	0.338	0.369	0.440	0.406	0.393	0.427	0.429	0.399	0.361	0.341	0.311		
	taud		2.159	2.089	2.150	2.094	2.273	2.331	2.254	2.231	2.275	2.350	2.276	2.213		
	Ebn at Noon		216	240	259	252	269	274	261	253	247	234	203	194		
	Edh at Noon		24	34	38	46	40	38	41	40	34	26	22	20		
(44) All-Sky Solar Radiation	RadAvg		274	548	962	1347	1805	1911	1829	1606	1065	580	237	171		
	RadStd		54	87	154	121	156	160	114	125	135	93	43	44		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+1.35	N/A	N/A	+1.67	+0.94	N/A	-274	-380	+209	+90
Regional (7 neighbors)	+1.26	N/A	N/A	+1.76	N/A	N/A	-248	-356	+205	+99

Nomenclature: See separate page